High-dimensional Deterministic Chaos in Spike Radio Emissions

Mayte Alvarez Trimino *and Adolfo L. Mendez, Ph.D.*
Miami Dade College, Padron Campus, Miami, FL

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Miami Dade College

Introduction

- Solar radio spikes are millisecond, high-energy bursts embedded in solar flares.
- **The origin and physical mechanism (chaotic deterministic vs. stochastic) of these spikes is unclear.**
- Analysis focuses on a strong flare event from Active Region 0486 (Oct 2003) with rich spike activity (Pick et al., 2005).
- **Objective:** Investigate spike dynamics for deterministic signatures using nonlinear methods.

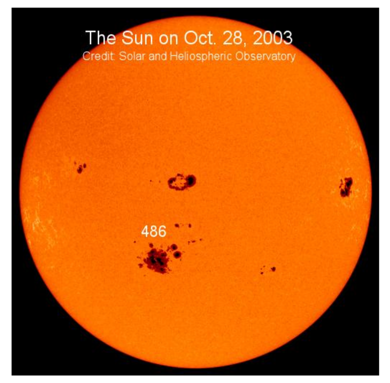

Fig. 1 Solar image of Active Region 0486 on October 28, 2003, credit SOHO.

Methodology

- High-resolution time series data analyzed (1,000 data points per second), recorded by the Trieste Astronomical Observatory at fixed frequencies: 237 MHz, 327 MHz, and 408 MHz.
- One optimal 4,000-point interval (10:00:10-10:00:14 UT) was selected for each frequency.

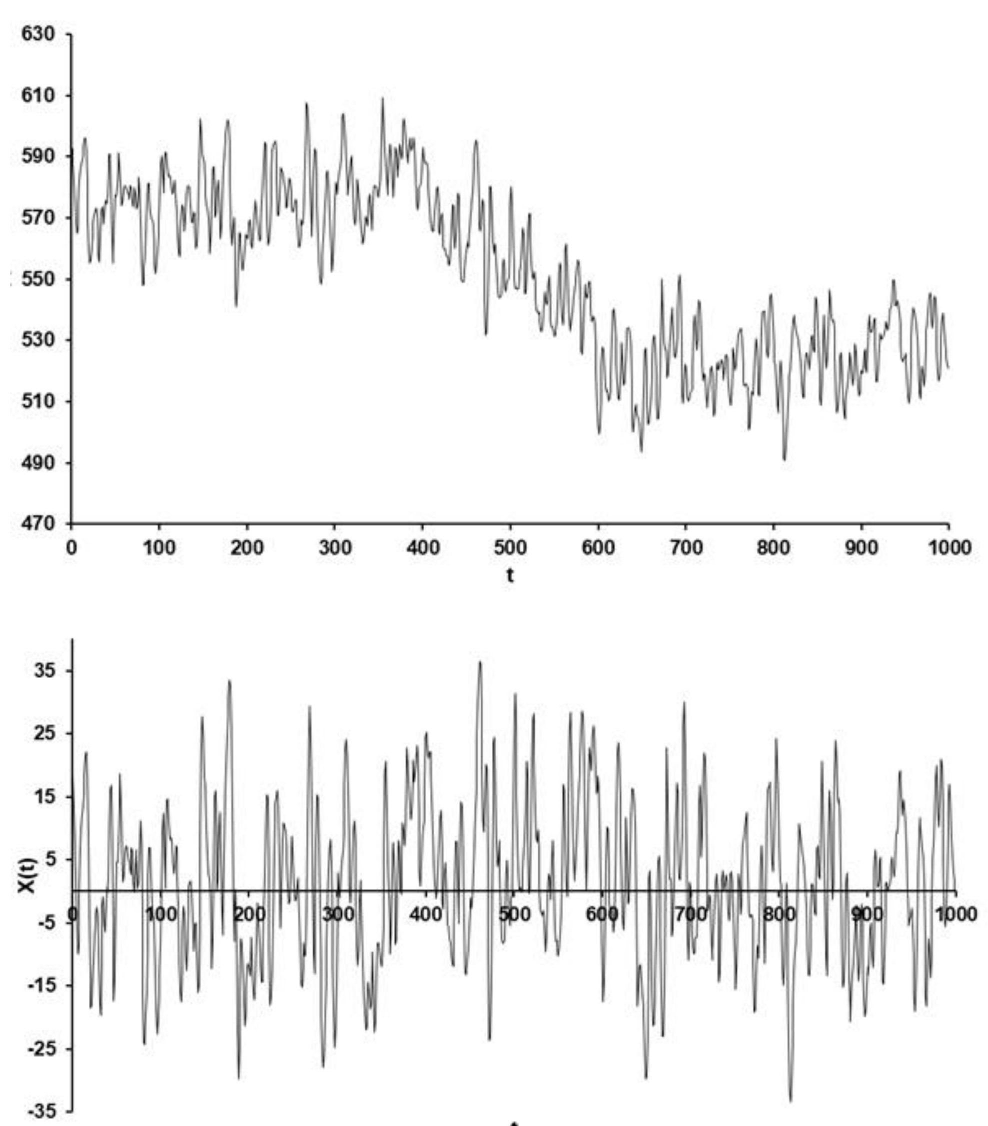

Fig. 2 Time profile of the fine structure at 237 MHz with a time resolution 1000 measurements per second **(a)** original signal, and **(b)** background-removed signal

- Time-delay embedding applied (Takens, 1981) to reconstruct the phase space using:

$$\vec{X}(t) = [x(t), x(t+\tau), \ldots, x(t+(m-1)\tau)]$$

Results

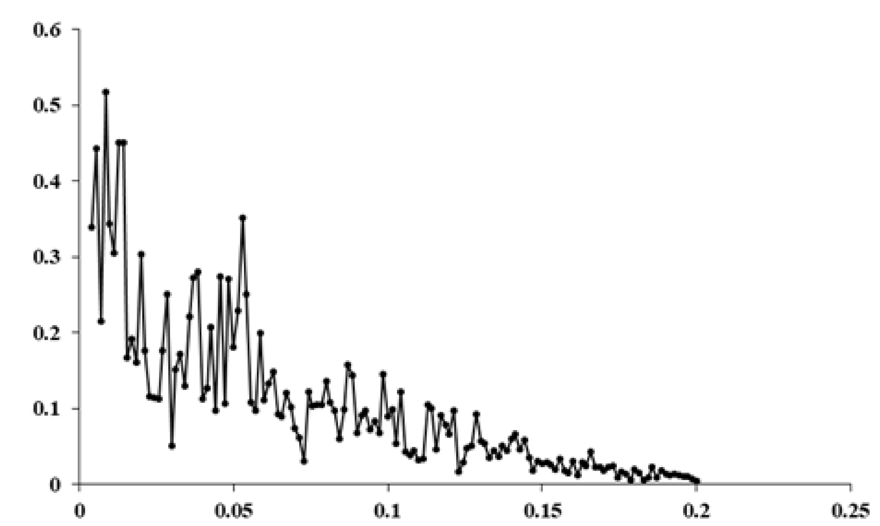

Fig. 3 Figure 4. Power spectrum of the 237 MHz spike emission obtained using the Fast Fourier Transform (FFT). Continuous broadband distribution without dominant peaks suggests a **lack of strict periodicity** and **implies nonlinear, chaotic dynamics.**

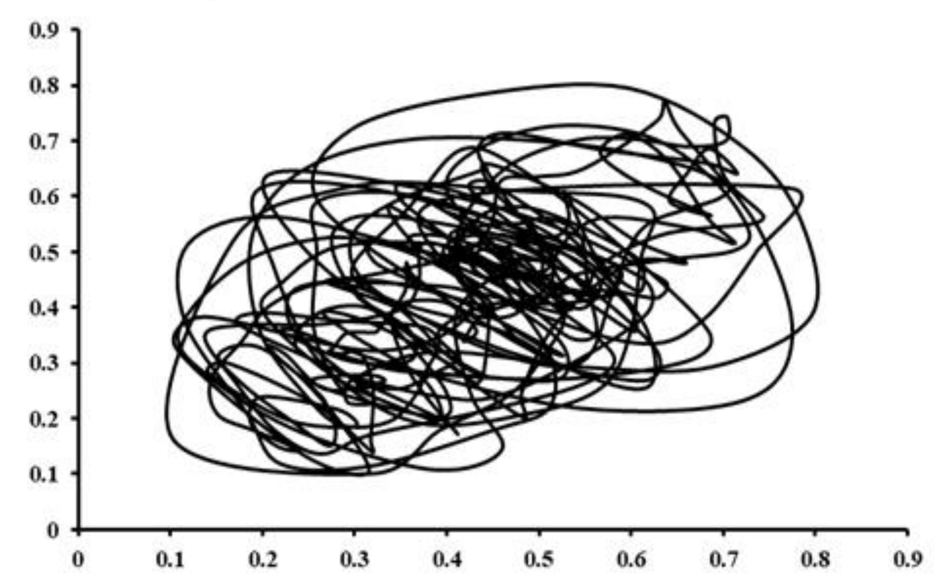

Fig. 4 2D state-space was reconstructed at 237 MHz using a time delay $\tau = 4$, determined from the first minimum of average mutual information (Frasser and Swinney, 1986).

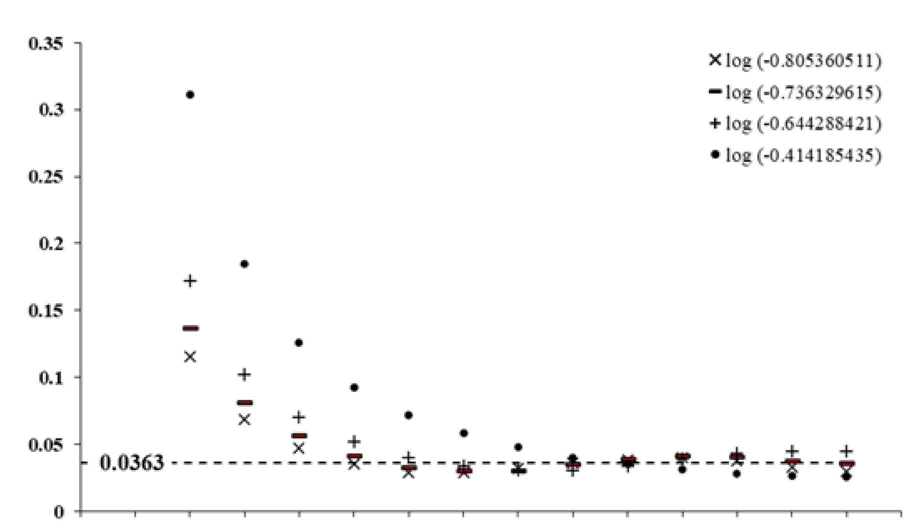

Fig. 5 Kolmogorov entropy (K_2) for the 237 MHz spike time series. Each symbol represents the value of log C(r) used in the entropy calculation at successive embedding dimensions. The average entropy $K_2 = 0.036 \pm 0.005$ denotes a finite information-production rate typical of deterministic chaotic systems, confirming that the signal is neither periodic nor purely stochastic (Grassberger, 1983).

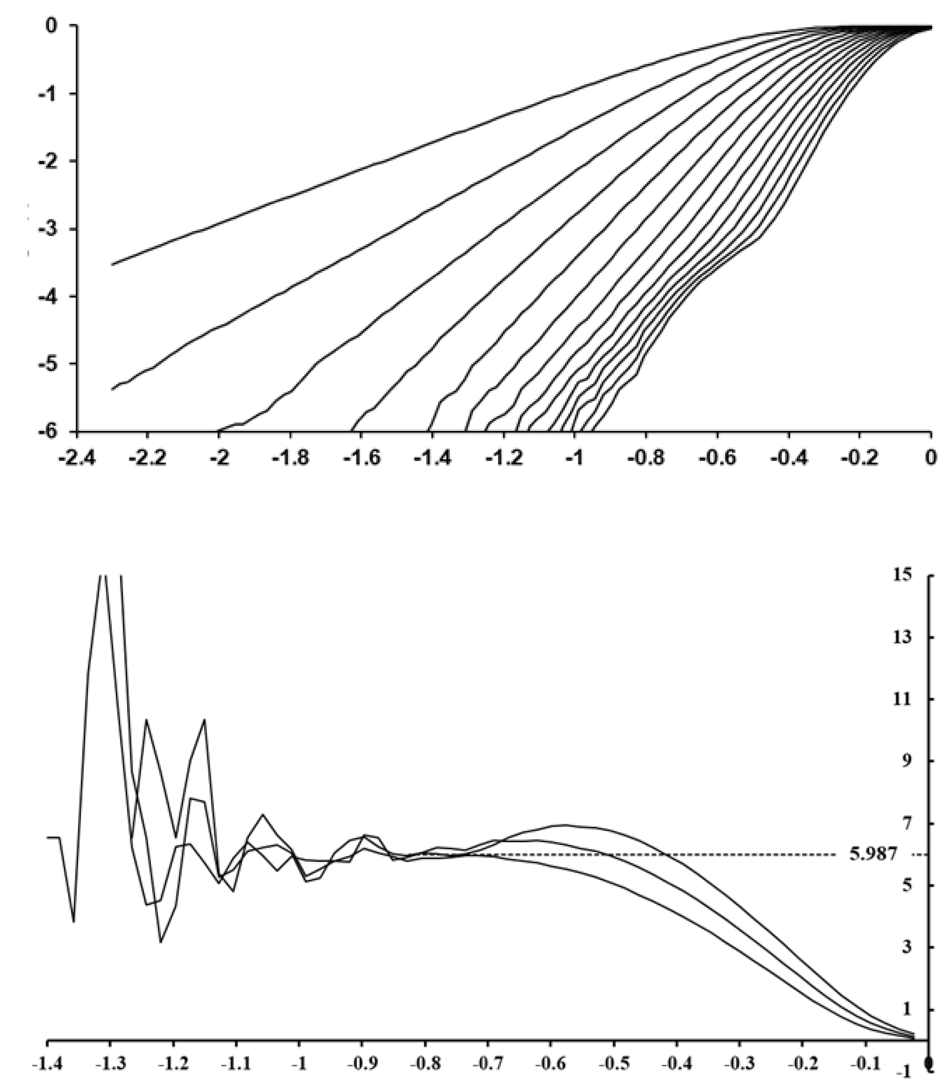

Fig. 6 **(a)** Correlation integral log [$C_d^{(2)}$] vs. log [r] for the time series at 237 MHz **(b)** Slopes of the linear scaling regions for 237 MHz (Isliker 1992).

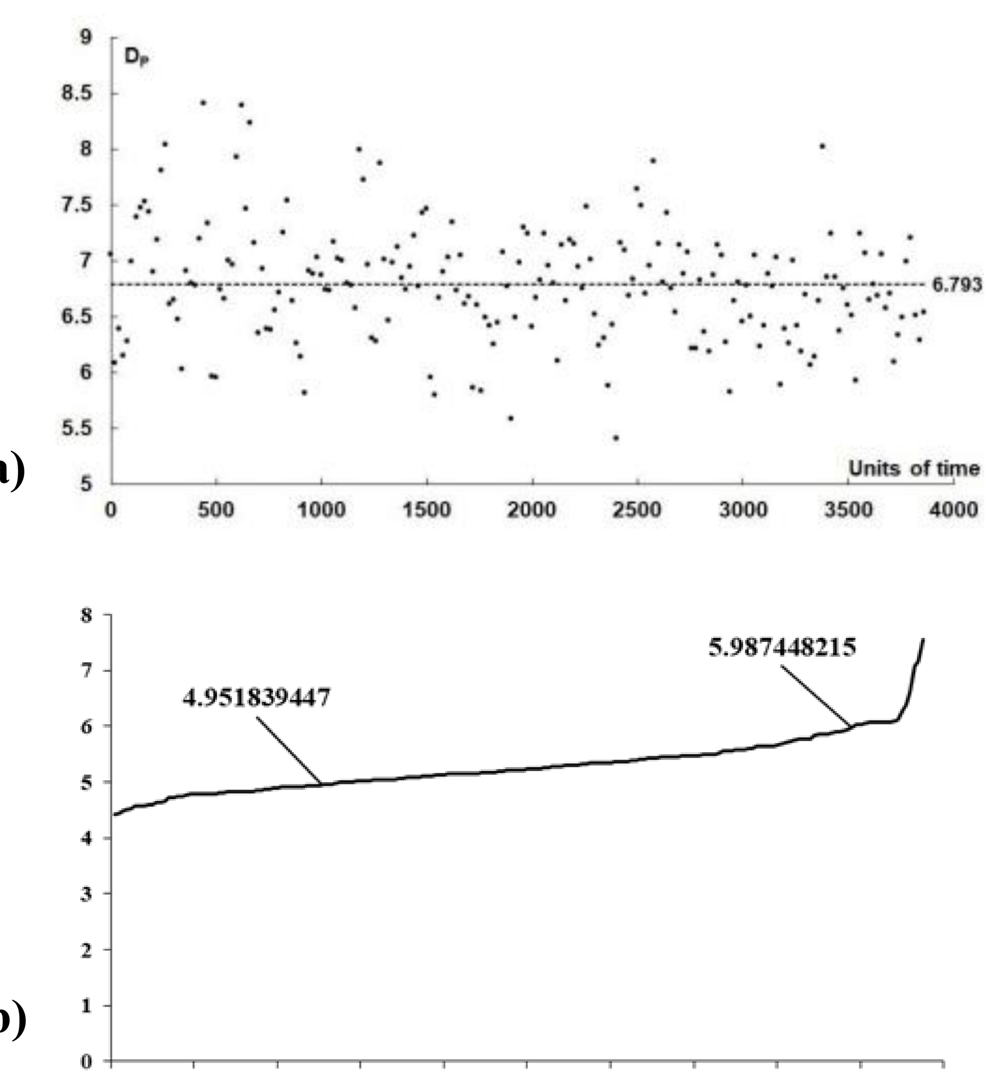

Fig. 7 Pointwise dimension (D_p) for the 237 MHz time series. (a) Local D_p shows fluctuations between ≈ 5.5 and 8.4, with an average $D_p \approx 6.793$. (b) Average D_p exhibits a stable linear scaling (70% of datapoints) with embedding dimension.

Conclusions

Frequency	λ	Δ	H	D_C	D_P
237 MHz	0.449	0.708	0.267	6.019±0.114	6.793±0.524
327 MHz	0.279	0.678	0.257	6.097±0.129	6.964±0.558
408 MHz	0.560	0.623	0.263	6.055±0.100	6.733±0.477

Table 1. Lyapunov exponent **λ**, Deterministic factor **Δ**, Hurst exponent **H**, Correlation dimension $\mathbf{D_C}$, and Pointwise dimension $\mathbf{D_P}$

- High Deterministic Factor across frequencies confirms deterministic dynamics.
- Positive Lyapunov exponents and Kolmogorov entropy confirm a chaotic deterministic behavior.
- Hurst exponent values indicate an anti-persistent turbulent process characteristic of non-regular time series.
- Calculated both correlation and pointwise dimension (D ≈ 6) is notably larger than those for fibers and pulsations refer to Mendez et al. 2013, 2015.
- No remarkable difference between the dimension across frequencies. This is evidence pointing to the difference between spikes and other fine structures bursts **generation mechanism**.
- The evolution of the spikes can be characterized by a **high-dimensional deterministic chaotic process**.

References & Acknowledgments

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